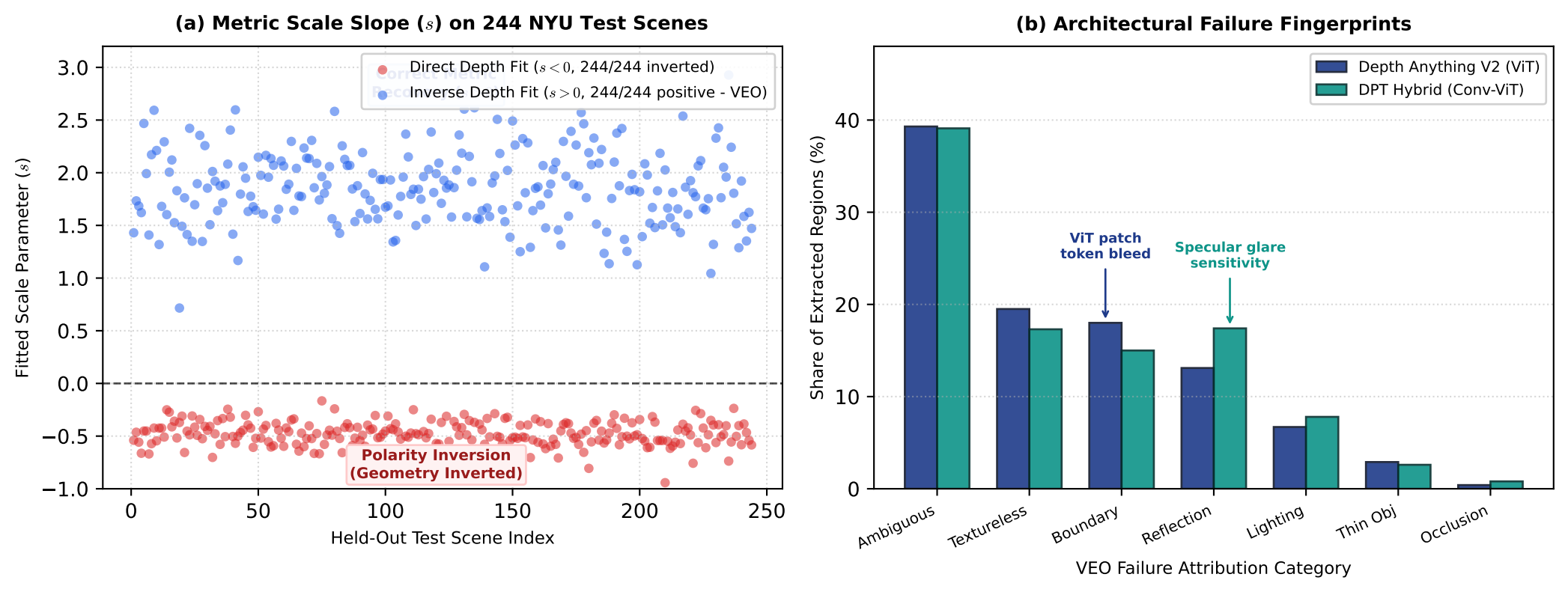
(a) Metric Scale Slope ($s$) on 244 NYU Test Scenes
Direct Depth Fit ($s < 0$, 244/244 inverted)
Inverse Depth Fit ($s > 0$, 244/244 positive - VEO)
Polarity Inversion
(Geometry Inverted)
Fitted Scale Parameter ($s$)
Held-Out Test Scene Index
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
50
100
150
200
250
(b) Architectural Failure Fingerprints
Depth Anything V2 (ViT)
DPT Hybrid (Conv-ViT)
ViT patch
token bleed
Specular glare
sensitivity
Share of Extracted Regions (%)
40
30
20
10
0
Ambiguous
Textureless
Boundary
Reflection
Lighting
Thin Obj
Occlusion
VEO Failure Attribution Category